

5633032

B.Tech. DEGREE EXAMINATION, JANUARY 2022.

Third Semester

Computer Science Engineering

DATA STRUCTURES

Time : Three hours

Maximum : 75 marks

PART A – (10 × 2 = 20 marks)

Answer ALL questions.

1. Define linear data structure. Give any two examples.
2. How to represent a two-dimensional array?
3. Define Dequeue.
4. What is ...

9. What is meant by symbol table?

10. List any four external storage devices.

PART B - (5 × 11 = 55 marks)

Answer ALL questions.

UNIT I

11. Write an algorithm, based on Fibonacci search, to search a number from a set of numbers.

Or

12. Illustrate different types of arrays with examples.

UNIT II

13. Discuss the functions of Circular linked list and its structure with a neat diagram.

Or

14. Describe the various operations on Stacks.

UNIT III

15. Explain the terminology of Tree, and illustrate the binary trees with neat structures.

Or

16. Write a detailed note on Trie tree indexing.

UNIT IV

17. List the differences between the Tree and Graph.

Or

18. Write a detailed note on Kruskal's algorithm.

UNIT V

19. Discuss various features of the Hash table.

Or

20. Discuss the following.

(a) Sequential file organisation

(b) Indexed file organisation.